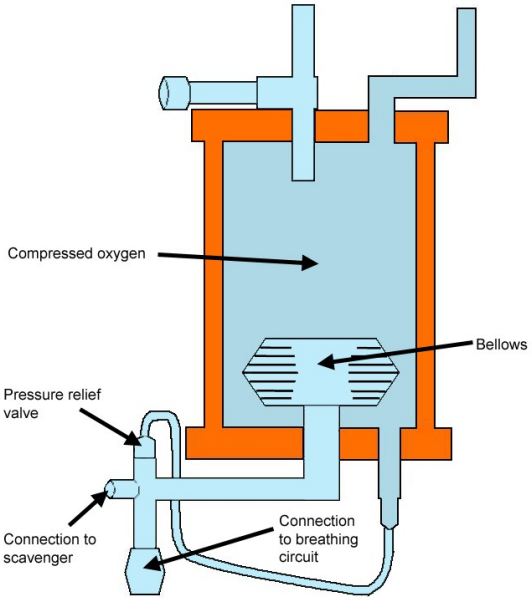
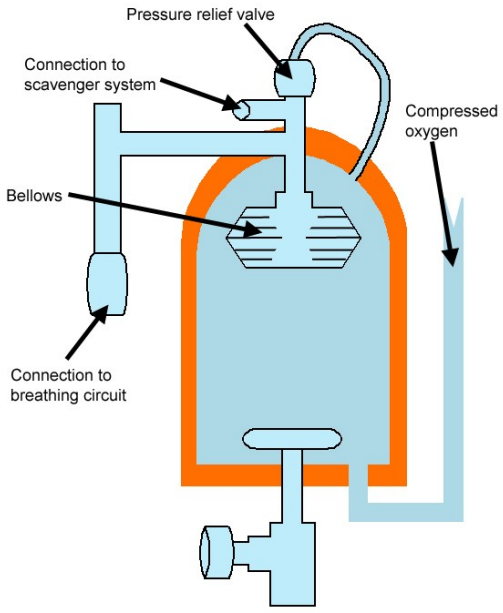
The bellows are an accordion-shaped bag contained within a cylinder. The bellows are filled with anesthetic gases that are delivered to patient when the bellows compress. The cylinder is clear so that the movement of the bellows can be easily monitored. There are two types of bellows: ascending and descending. They are named according to the direction of movement during the exhalation phase of the ventilatory cycle.

Ascending bellows are attached to the base of the cylinder and rise during the exhalation phase. Gases exhaled from the patient fill the bellows. During the inspiratory phase, compressed oxygen surrounding the bellows drive the bellows downward and force the anesthetic gases toward the patient. If there is a disconnection in the anesthesia circuit between the bellows and the patient, the bellows will fail to rise, giving the anesthetist an immediate indication that the disconnect occurred. This feature also provides about 2-3 cm H_2O of positive-end expiratory pressure to the patient so that the bellows will rise during exhalation.



Ascending Bellows



Descending (Hanging) Bellows

Descending bellows are attached to the top of the cylinder. The bellows fall during exhalation and rise during the inhalation phase. The bottom plate of the bellows is usually weighted to facilitate its fall during exhalation. If a disconnection occurs in the circuit between the bellows and the patient, the bellows will still fall. Not only does this not alert the anesthetist of the disconnection, but the fall of the bellows will entrain room air into the circuit through the leak or disconnection. Due to the high risk associated with this, most modern anesthesia machines incorporated ascending bellows for added safety. Some newer models have begun to utilize descending bellows again, but employ built-in CO_2 monitors and additional sensors and alarms for increased safety.

Sandberg W, Urman RD, and Ehrenfield JM. The MGH Textbook of Anesthetic Equipment. Philadelphia, PA: Elsevier; 2011: 54.

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